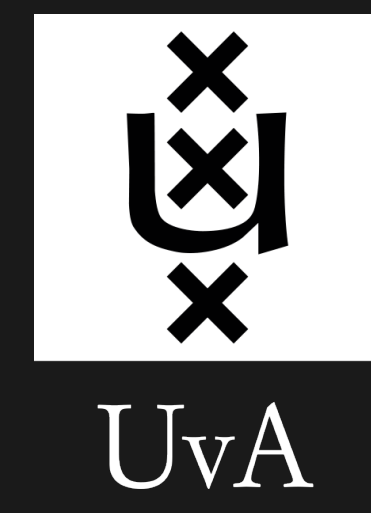


ARTHRITIS HAND BRACE



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Introduction

Arthritis is a disease that affects the joints and causes pain [1]. Arthritis braces support and relieve the affected joints [2]. We set out to develop a custom inflatable brace to tackle some of the drawbacks of currently available custom braces.

- **Current options** can be expensive. They require a high degree of specialization and suffer from itchiness and poor ventilation.
- **Inflatable braces** are cheap, simple to manufacture, and easy to customize, which makes them ideal for use in places without access to advanced equipment.

Methodology

- We used **Nylon** coated with **TPU** on one side. This material can be sealed using a 3D printer to create cells that can be filled with **air**. We experimented with numerous prototypes as well as different cell structures, such as hexagons and triangles, before settling on mostly rectangular shapes due to their relatively high volume, offering more support.
- We developed a blender program that generates a 3D model of a brace based on just **five** measurements for **accessibility**.
- We analyzed the shrinkage factor (width inflated / width deflated, corrected for seal lines) of 10x10 cm samples with multiple cells (see **Figure 1**) to obtain insight into how the width of each cell would affect the brace when inflated. To do this, we inflated 10x10 cm samples to a pressure of 0.3 bar, as that is a safe but firm enough pressure for a brace.

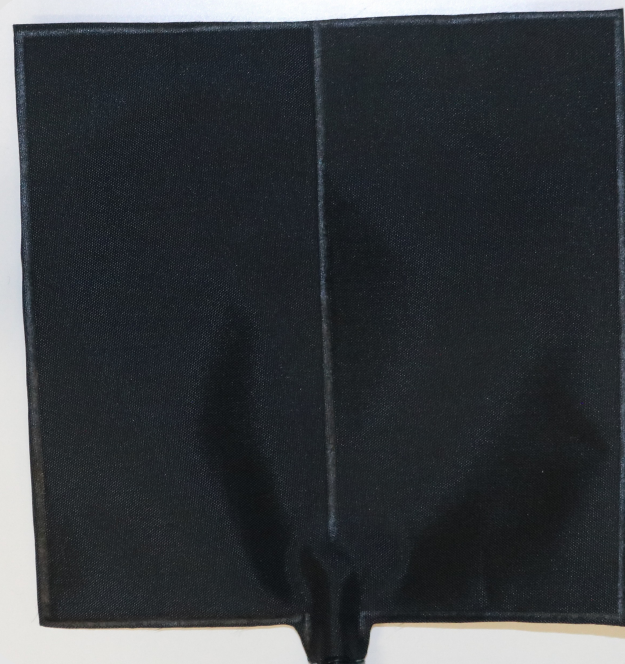
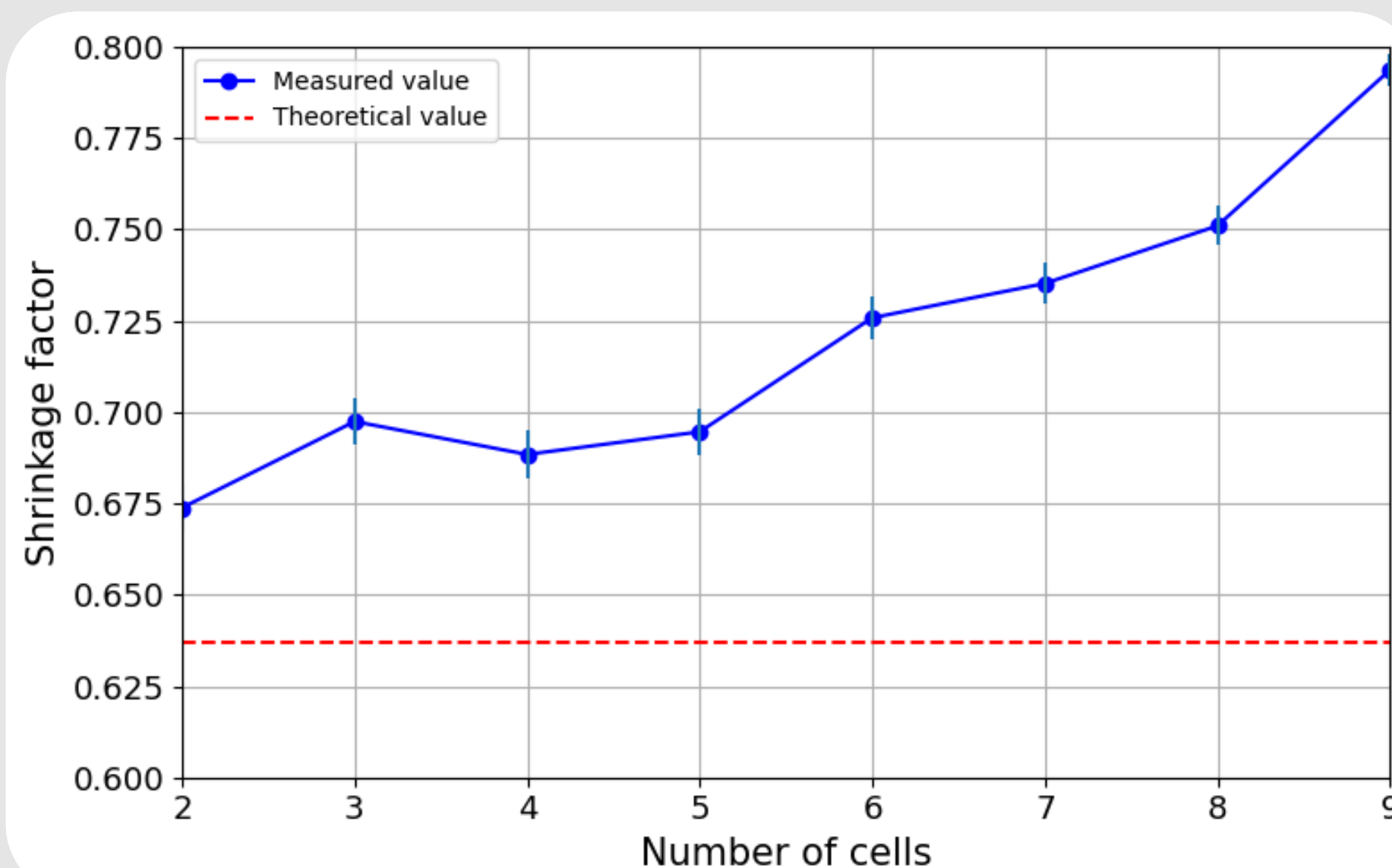


Figure 1: sample with two cells

Discussion & Conclusion

- This process can be **developed further**, most notably the generative blender program and the cell structure of the brace. The shrinkage ratios of the cells could be tested more thoroughly, which would allow them to be incorporated into the design to compensate for the change in size when inflated.
- The current cell structure is designed to support the thumb joint, as well as the rest of the hand, but without **input from experts**, this part of the design remains subjective.
- The material has a limit in how sharp the creases can be. This radius gets relatively bigger as the cells get smaller, which results in a deviation from the theoretical value. A small dip is visible between 3 and 5 cells. While this does not make physical sense, it could be explained by a measurement error.
- To conclude, this project is meant to serve as a **proof of concept** of an inflatable arthritis hand brace that is affordable and accessible. We hope that this project will pave the way for further development of inflatable braces.

Results



Cell shrinkage factor test results

The upward trend indicates a negative linear correlation between cell width and the shrinkage factor. Tests were performed on 10x10 cm cell samples at 0.3 bar. The theoretical value of $2/\pi$ is based on the cross section of each cell forming a perfect circle.

Figure 2: shrinkage factor per number of cells

- **We developed a process** to make a **tailored** inflatable arthritis hand brace without the need for advanced tools such as 3D scanners.

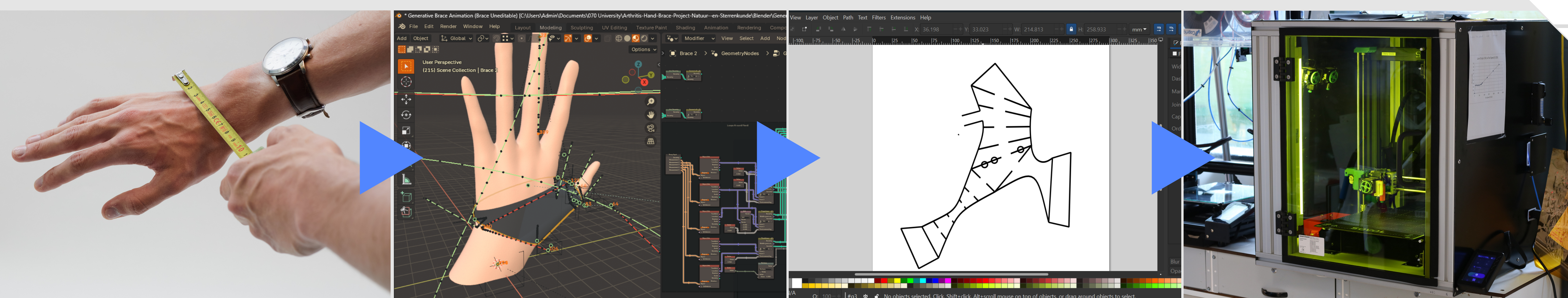


Figure 3: our process

- **5 Measurements** → **3D model** → **2D vector image** → **3D printer sealing** → Cut brace out → Insert tube and **inflate**

